PPP-D-723J
June 17, 1988
SUPERSEDING
PPP-D-723H
September 16, 1980

FEDERAL SPECIFICATION

DRUMS, FIBER

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers new fiber drums for use in domestic shipment, normal overseas shipment, and military overseas shipment.

1.2 Classification.

1.2.1 Part number. The part number shall be a combination of numbers and letters in accordance with the following:

PPPD723 - 1 B 4 E 7 A C C

- 7 A C C — Dash number to represent requirements as stated in the specification
- 4 — Dash number, class (4 = Class 4)
- E — Dash number, grade (B = Grade B)
- 1 — Dash number, type (1 = Type I)

1.2.2 Types, grades, and classes. Fiber drums covered by this specification shall be of the following types, grades, and classes, as specified (see 6.2).

Type I - Domestic (nonweather resistant), see 6.1.1
Type II - Overseas (nonweather resistant), see 6.1.2 and 6.1.4
Type III - Overseas (weather resistant), see 6.1.3 and 6.1.4

Grade A - For dry and solid material (applicable to all types)
Grade B - For semiliquid material (applicable to all types)
Grade C - For hot poured materials that solidify on cooling (applicable to types I and II only)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8110

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Grade D - For rolled or cylindrical items (applicable to types I and II only) see 6.1.1.1
Grade E - For liquids or articles in liquid - nonregulated (applicable to all types)

Class 1 - Regular construction (applicable to only types I and III grade A drums)
Class 2 - Foil laminated construction (applicable to only types I and III grade A drums)
Class 3 - Integral plastic lining (applicable to only grade E drums)
Class 4 - Semi-rigid plastic component (open head loose liner) applicable to only grade E drums
Class 5 - Molded rigid one-piece plastic component (closed head liner) applicable to only grade E drums

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Federal Standards:

FED-STD-101 - Test Procedures for Packaging Materials
FED-STD-123 - Marking for Shipment (Civil Agencies)

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Single copies of this specification and other Federal specifications, standards, and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.)

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(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

Military Standards:

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-147	- Palletized Unit Loads
MIL-STD-731	- Quality of Wood Members for Containers and Pallets

(Copies of military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting activity.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

Federal Regulations:

21 CFR 100-1299	- Federal Food, Drug, and Cosmetic Act
49 CFR 100-199	- Transportation

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

American Society for Testing and Materials (ASTM) Standards:

D 87	- Melting Point of Petroleum Wax (Cooling Curve)
D 256	- Impact Resistance of Plastics and Electrical Insulating Materials
D 638	- Tensile Properties of Plastics
D 648	- Deflection Temperature of Plastics Under Flexural Load

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D 1008	- Water Vapor Transmission of Shipping Containers
D 1238	- Flow Rates of Thermoplastics by Extrusion Plastometer
D 1505	- Density of Plastics by the Density-Gradient Technique
D 1525	- Test Method for Vicat Softening Temperature of Plastics
D 1693	- Environmental Stress-Cracking of Ethylene Plastics
D 3951	- Standard Practice for Commercial Packaging

(Copies should be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadephia, PA 19103-1187.)

American Iron and Steel Institute Standard:

Steel Product Manual, Sheet Steel

(Copies should be obtained from Iron and Steel Society, P.O. Box 411, Warrendale, PA 15086.)

Technical Association of the Pulp and Paper Industry (TAPPI) Standard:

T 803 M 50 - Puncture and Stiffness Test of Container Board

(Copies should be obtained from the Technical Association of the Pulp and Paper Industry, Technology Park Atlanta, P.O. Box 105113, Atlanta, GA 30348.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.4).

3.2 Materials. Materials not definitely specified shall be of the quality normally used by the manufacturer for fiber drums provided that the complete item complies with all provisions of this specification.

3.2.1 Sheet steel. Unless otherwise specified herein, steel shall be in accordance with the required "Manufacturer's Standard Gage Number" of the Steel Product Manual, Sheet Steel; low carbon; open hearth; cold rolled steel. Individual steel sheets shall be subject to the "Manufacturer's Standard Gage Number" covering variations and tolerances, as specified in table I.

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TABLE I. Variations and tolerances of gages

Gage No.	Nominal thickness 1/ (inches)	Minimum thickness 1/ (inches)
32	0.0097	0.0094
30	0.0120	0.0110
28	0.0149	0.0129
26	0.0179	0.0159
24	0.0239	0.0209

1/ Thickness shall be measured at any point on the sheet not less than 3/8 inch from any edge.

3.2.2 Reclaimed fiber. Unless otherwise specified (see 6.2), the materials used to manufacture fiber drums shall contain a minimum of 5 percent reclaimed fibers from the sources described in 6.5. Unless otherwise specified, no maximum percent is set provided the physical characteristics meet the specified requirements. The use of reclaimed fibers shall not constitute a waiver of the specification requirements.

3.2.3 Food and drug containers. All materials used in the manufacture of fiber drums intended for contact with food and drugs for human or animal consumption must conform to rules and regulations set forth in 21 CFR 100-199.

3.3 Construction of type I (see 6.1.1).

3.3.1 Type I, grade A, class 1. Each drum shall be constructed with the fiberboard side wall joined to top and bottom heads (see 3.3.1.3). The side wall shall be either single or double convolutely wound and as specified in table II for the applicable weight of contents or capacity as specified (see 6.2). The heads shall be either steel, wood, fiberboard, plywood, plastic, or a combination of steel and paperboard at the option of the contractor unless otherwise specified (see 3.6 and 6.2), and have requirements specified in table II for the applicable weights of contents or capacity. The finished drum shall meet the performance requirements of 3.8.1.1.

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TABLE II. Side wall and heads for type I, grade A, class 1

Maximum content weight (lb). 1/	Maximum capacity (gal)	Minimum side wall bursting strength (psi) 2/	Heads, top and bottom 4/				
			Fiberboard 3/				
			Minimum thickness (inches)	Minimum bursting strength (psi)	Minimum steel gage	Minimum solid wood thickness (inches)	Minimum plywood 3 or more plies (inches)
60	30	400	0.120 or 0.090	300 or 600	30	1/2	3/11
115	45	500	0.160 or 0.120	400 or 800	28	1/2	3/10
150	55	600	0.160 or 0.120	400 or 800	28	1/2	3/8
225	65	700	0.180 or 0.120	500 or 1000	26 5/	1/2 7/	3/8
300	75	800	0.200 or 0.160	550 or 1100	26 5/	25/32	7/16
400	75	900	0.240 or 0.200	600 or 1200	24 6/	25/32	7/16
550	75	1000	0.220	1300	24 6/	25/32	7/16

1/ The minimum requirements for side wall and headings shall be governed by the greater requirement of weight of contents or capacity.

2/ Cady or Mullen testing method for fiberboard components (see 4.4.3): Either of the test methods (see 4.5.1.1) may be used. When more than single ply, test shall be determined from the summation of the tests of individual plies; when test is made on a completed drum, the punctures shall be made from the exterior to the interior surface, in which case

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the values for side wall shall be not less than 80 percent of the value in the above table. The average of the number of unit findings specified in table XX shall not be less than the prescribed minimum requirements.

3/ When headings are made of two or more discs, they shall be firmly glued together. Thickness may be either dimension for listed capacities, but bursting strength must compensate by meeting corresponding psi minimum requirements.

Example: 30 gallon drum headings may be made with either a 0.120 inch thickness minimum and a 300 psi bursting strength minimum or a 0.090 inch thickness minimum and a 600 psi bursting strength minimum.

4/ Plastic, when used, must be constructed of injection molded high density polyethylene or polypropylene and have a minimum thickness of 0.090 inch. Material must meet the following parameters as specified within the referenced American Society for Testing and Materials (ASTM) testing methods:

a. Injection molded high density polyethylene homopolymers or copolymers -

Density, ASTM D 1505	0.948 to 0.967
Melt index, ASTM D 1238 FR-E	1.0 to 8.0
Tensile strength, ASTM D 638	3000 psi, minimum
Heat resistance, Vicat Soft Point	240° to 260°F
Stress cracking, ASTM D 1693, Method B, F50	20 hours, minimum

b. Injection molded polypropylene homopolymers or copolymers -

Density, ASTM D 1505	0.890 to .915
Melt flow, ASTM D 1238	1.0 to 8.0
Tensile strength, ASTM D 638	3000 psi, minimum
Izod, impact, ASTM D 256, notched at 23°C	1.0 to 18.0
Heat distortion temperature at 66 psi, ASTM D 648	160° to 275°

5/ Bottom may be constructed of not less than 30 gage steel combined with paperboard having a minimum thickness of 0.110 inch and Mullen test of not less than 400 psi.

6/ Bottom may be constructed of not less than 30 gage steel combined with paperboard having a minimum thickness of 0.140 inch and Mullen test of not less than 550 psi.

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7/ Joints in heads shall be Linderman joints, tight and firmly glued with water-resistant glue, except as specified for kraft covered heads herein. When it is known that the weight of content exceeds 200 pounds and when the fiber shell of drum is provided with a formed inside bead to support the wood head and a steel rim holds the head in place and is locked in an external groove of the fiber shell, solid wood heads shall be not less than 1/2 inch thick. Such 1/2 inch heads shall be reinforced by covering both sides with kraft liner board not less than 0.012 inch thick securely glued throughout entire area of contact with a water resistant glue. A joint of any type shall be permitted for kraft covered heads.

3.3.1.1 Single convolutely wound unit. Fiberboard used in the side wall shall be convolutely wound with no single ply less than 0.012 inch thick. All plies shall be firmly glued together.

3.3.1.2 Two convolutely wound units. The side wall shall consist of two cylindrical units glued together or one closely fitted within the other. Each unit shall be convolutely wound with no single ply of fiberboard less than 0.012 inch thick and each unit shall have all plies firmly glued together. The inner unit may be fabricated from nontest fiberboard. However, the combination of the two units shall meet the requirements of table II for the applicable weights of contents or capacity.

3.3.1.3 Methods of fastening top and bottom headings to side wall. Headings of drums shall be fastened to the side walls with adhesive, metal fasteners, lock stitch thread, or tape, or a combination of these methods. When metal fasteners are used, they shall be clean, have a smooth finish, and be free of sharp edges. When tops and bottoms are made of fiberboard and provided with wood segments reinforcing the circumference, the wood segments shall be minimum 1/2 inch thick and 1-1/4 inches wide at the joint between segments. The fiberboard shall then be not less than the bursting strength specified for the side wall.

3.3.1.4 Barrier, lining, or interior coating. When specified (see 6.2 and 6.3), the drums shall be furnished with a barrier, lining, or interior coating.

3.3.2 Type I, grade A, class 2. Each drum shall have side wall and headings joined so as to meet the requirements of 3.8.1.2 and 3.8.3. The side wall construction shall be as specified in 3.3.2.1.

3.3.2.1 Side wall construction. Unless otherwise specified (see 3.3.2.1.1), the side wall shall be a single convolutely wound unit using sodium silicate adhesive or equivalent with no single ply less than 0.012 inch. The construction of the side wall shall consist of 8 plies from inside to outside of the drum, of which the first or second inside plies shall be an

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aluminum barrier sheet containing minimum 0.00035 inch aluminum foil. The ply incorporating the foil shall consist of a three-layer structure of kraft-foil-kraft or polyethylene-foil-kraft (see 6.2). The ply shall have a thickness of 0.12 inch.

a. If sheet is kraft faced on both sides, the sheet must have a 4-inch overlap.

b. If sheet is polyethylene coated on one side, it must be the inside ply. Butt seam joint with a heatseal tape or pressure-sensitive plastic tape over the joint.

NOTE: All paperboard and laminated weights are given on a ream basis of 24 inches by 36 inches - 500 sheets.

3.3.2.1.1 Exception. When specified (see 6.1 and 6.2), side wall construction shall be as specified for 550 pounds, except that when gallonage capacity does not exceed 55 gallons, interior side wall and bottom of drum shall be laminated with not less than 0.001 inch thick aluminum foil and provide a ground from the interior surface of the drum side walls to the outer surface of the bottom. The net weight of contents shall not exceed 600 pounds.

3.3.2.2 Bottom. The drum bottom shall have the same aluminum sheet for the inside surface as is used for the sidewall. Total bottom thickness shall be not less than 0.160 inch with a minimum dry bursting strength of 900 psi.

3.3.2.2.1 Method of fastening bottom to side wall. The bottom shall be fastened to the side wall as specified in 3.3.1.3.

3.3.2.3 Cover. The drums shall have a full removable cover made from 26 gage U.S. manufacturer's standard cold rolled steel. After fabrication, the cover shall be degreased and given a protective coating of baked alkyd varnish.

3.3.2.3.1 Cover gasket. The cover shall be fitted with either a 1/4-inch diameter round or tubular sponge rubber gasket cemented in place or a vulcanized flowed-in gasket. The flowed-in vulcanized gasket shall be of cold process Buna S latex extended with inorganic fillers and shall weigh not less than 6 grams per foot.

3.3.2.3.2 Locking band. The cover shall be fastened to the drum by means of a lever actuated locking band. The locking band shall be formed from electrogalvanized or galvanized steel having a minimum thickness of 0.022 inch. The handle and lock shall be formed from minimum 0.047 inch cold rolled steel and shall be varnished or plated. The clamping ring safety latch shall be so constructed that a wire and lead seal may be affixed after closure in such a manner that the cover cannot be removed without destroying the seal.

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3.3.2.3.3 Chimes. The top chime used for fastening the cover and locking ring to the side wall and the bottom chime used for fastening bottom to side wall shall be made of galvanized or electrogalvanized steel having a minimum thickness of 0.022 inch. Chime bands shall be made from flat or preformed stock, joined together by welding. The weld may be butt or lap at the option of the contractor. The chime band shall be die crimped or spun into position to provide the water vapor transmission protection specified in 3.8.3.

3.3.2.4 Size and content. Except as specified in 3.3.2.1.1, the size of the drums shall be as specified by the procuring activity (see 6.2). The net weight of contents shall be a maximum of 200 pounds.

3.3.2.5 Interior treatment. When the aluminum foil is used for other than a lining (see 3.3.2.1 and 3.3.2.2), the drum interior shall be wax impregnated using a fully refined paraffin having a minimum melting point of 125°F when tested as specified in 4.5.7.

3.3.3 Type I, grade B drums for semiliquid material (see 6.1). Drums shall consist of side walls and headings and shall conform to the requirements of table III for size and capacity as specified (see 6.2).

3.3.3.1 Side wall construction. The method of fastening top and bottom to side wall shall be as specified in 3.3.1.1 through 3.3.1.3.

TABLE III. Side wall and heads for type I, grade B

Maximum content weight (lb) 1/	Maximum capacity (gal)	Minimum side wall bursting strength (psi) 2/	Heads, top and bottom 4/				
			Fiberboard 3/				
			Minimum thickness (inches)	Minimum bursting strength (psi)	Minimum steel gage	Minimum solid wood thickness (inches)	Minimum plywood 3 or more plies (inches)
60	10	500	0.120 or 0.090	300 or 600	30	1/2	3/11
115	20	600	0.160 or 0.120	400 or 800	28	1/2	3/10
150	30	700	0.160 or 0.120	400 or 800	28	1/2	3/8

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TABLE III. Side wall and heads for type I, grade B (cont'd)

Maximum content weight (lb) 1/	Maximum capacity (gal)	Minimum side wall bursting strength (psi) 2/	Heads, top and bottom 4/				
			Fiberboard 3/				
			Minimum thickness (inches)	Minimum bursting strength (psi)	Minimum steel gage	Minimum solid wood thickness (inches)	Minimum plywood 3 or more plies (inches)
225	40	800	0.180 or 0.120	500 or 1000	26 5/	1/2 7/	3/8
300	50	900	0.200 or 0.160	550 or 1100	26 5/	25/32	7/16
400	60	1000	0.240 or 0.200	600 or 1200	24 6/	25/32	7/16
550	75	1100	0.220	1300	24 6/	25/32	7/16

1/, 2/, 3/, 4/, 5/, 6/, 7/ See footnotes to table II.

3.3.3.2 Barrier, lining or interior coating. The drum must be so treated, processed or proofed, or equipped with a plastic film bag type liner as to prevent absorption of contents.

3.3.4 Type I, grade C drums for hot-poured materials that solidify on cooling (see 6.1.4 and 6.2). Unless otherwise specified, drums shall consist of side walls and headings and shall conform to the requirements of table IV for size and capacity as specified (see 6.2). When specified (see 6.2), drums for hot-poured material having a minimum melting point of 160°F that solidify on cooling shall be packed in a drum having side walls made from three plies of virgin kraft suitably sized to caliper not less than 0.041 inch and have a Mullen test of not less than 400 psi. The bottom shall be minimum 33 gage metal crimped so as to prevent leakage of molten products. The inner ply of the side wall and top and bottom head shall be release coated. The top shall be secured as specified in 3.3.1.3. Side wall shall be either single or double convolute wound or lap joined as specified in 3.3.4.1 or 3.3.4.2.

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3.3.4.1 Side wall construction. Side wall construction of single and two convolutely wound units shall be as specified in 3.3.1.1 and 3.3.1.2.

3.3.4.2 Lap joint. Lap joint of single side wall shall be joined by overlapping ends. The overlap shall be not less than 1 inch and shall be firmly glued together throughout the entire area of contact.

3.3.4.3 Top and bottom fastening. Method of fastening top and bottom to side wall and interior protection shall be as specified in 3.3.1.3.

3.3.5 Type I, grade D (see 3.5.2 and 6.1.3). Drums shall consist of side walls and headings and shall conform to the requirements specified in table V for size and capacity as specified (see 6.2).

3.3.5.1 Body side wall construction. The side wall shall be a single convolutely wound unit and shall consist of multiple plies of kraft or jute liner board with no ply less than 0.012 inch thick. All plies shall be glued together, and the outer ply shall be water resistant in accordance with commercial standards for fiber drums. Rosin sizing, or high alkaline resistant sizing, wax coating, or the application of shellac or clear lacquer to the outer ply will be acceptable. When applicable (see 6.2), side wall shall be a single spirally wound unit of plies specified above.

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TABLE IV. Sidewall, fiberboard heading and metal heading for type I, grade C 1/

Maximum content weight (lb)	Maximum capacity (gal)	Maximum inside diameter (in)	Melting point of packed material (°F)	Sidewall 2/ 3/		Fiberboard, top and bottom 2/ 3/		Metal, top and bottom		Fiberboard, flap top 3/	
				Min thick (in)	Min dry bursting strength (psi)	Min thick (in)	Min dry bursting strength (psi)	Min top thickness (gage)	Min bottom thickness (gage)	Min thick (in)	Min dry bursting strength (psi)
115	-	15	200	0.024	200	0.024	200	33	33	0.024	200
115	-	15	180	0.030	270	0.030	270	33	33	0.030	270
115	-	15	160	0.039	340	0.039	340	33	33	0.039	340
115	-	15	140	0.060	370	0.060	370	32	32	-	-
115	-	15	120	0.090	400	0.090	400	32	32	-	-
-	55	23	160	0.120	750	0.120	600	28	26	-	-
-	55	23	120	0.120	1000	0.120	900	28	24	-	-

1/ The minimum requirements in the above table for side wall, top and bottom are governed by either the weight of contents in first column or by the capacity in second column, whichever requirements are higher.

2/ Cady or Mullen testing method for fiber components (see 4.4.3): Either of the test methods may be used. When more than single ply, test shall be determined from the summation of the tests of individual plies; when test is made on a completed drum, the punctures shall be made from the exterior to the interior surface, in which case the values for side wall shall be not less than 80 percent of the value in the above table. The average of the number of unit findings specified in table XX shall be not less than the prescribed minimum requirements.

3/ For weights of contents of 115 pounds maximum and contents having a melting point of 160°F, the side walls, top and bottom may be 0.045 inch thick with a dry bursting strength of 200 pounds (see 3.3.4).

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TABLE V. Body and cover side wall and heading discs for types I and II, grade D drums

Types I and II	Inside diameter (inches)		
	Up to 11 incl.	12 to 14 incl.	Over 14 to 16 incl.
Minimum dry bursting strength, (psi) 1/:			
Body side wall	650	850	1000
Cover side wall	500	600	700
Top and bottom discs, fiberboard	850	1200	1500
Minimum dry puncture resistance (units) 2/:			
Body side wall	500	700	900
Cover side wall	400	500	600
Top and bottom discs	700	1050	1350
Minimum thickness of top and bottom heading discs (inches):			
Fiberboard	0.300	0.400	0.600
Wood	1/2	1/2	1/2
Plywood	3/8	7/16	7/16

1/ Cady or Mullen testing method for fiber components (see 4.4.3): Either of the test methods may be used. When more than single ply, test shall be determined from the summation of the tests of individual plies; or, when test is made on a completed drum, the puncture shall be made from the exterior to the interior surface, in which case, the values for side wall shall be not less than 80 percent of the value in the above table. The average of the number of unit findings specified in table XX shall be not less than the prescribed minimum requirements.

2/ See 4.4.3.

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3.3.5.2 Heading (top and bottom).

3.3.5.2.1 Discs. Discs shall be constructed of multiple plies of kraft or jute, or laminated solid fiberboard, plywood, or solid wood, having a combined thickness, as specified in table V. No ply of fiber or jute shall be less than 0.012 inch thick and the plies shall be glued together. The outer ply of the fiber disc shall be water resistant, as specified in 3.3.5.1. When wood discs are fabricated from more than a single piece of lumber, all joints shall be tight and glued.

3.3.5.2.2 Cover side wall. Cover side wall shall be a single convolutely wound unit and shall be a minimum of 4-3/4 inches in inside depth, and shall consist of multiple plies of kraft or jute liner board, with no ply less than 0.012 inch thick, glued together with the outer ply water resistant, as specified in 3.3.5.1. Cover side wall shall conform to the requirements in table V.

3.3.5.2.3 Tray. The tray shall be of kraft fiberboard having a minimum thickness of 0.025 inch.

3.3.5.2.4 Collar. The collar shall be of multiple plies of kraft or jute liner board laminated together. No ply shall be less than 0.012 inch thick. Total combined thickness of collar shall be not less than 0.280 inch. The width of the collar shall be not less than 5/8 inch or more than 1 inch. Alternatively, the collar may be a twisted kraft strip having a filler of 30 to 60 pounds basis weight paper and a wrapper of 30 to 40 pounds basis weight paper. Components shall be compressed to result in a strip measuring not less than 5/8 inch in width and 1/4 inch in thickness. Alternatively, the collar may be of a single thickness of lumber having minimum dimensions of not less than 5/8 inch in width by not less than 3/16 inch in thickness.

3.3.5.2.5 Drum end positioning and fastening. The disc shall be positioned in the tray and the collar shall be placed over the disc, within the inside perimeter tray. The assembly shall be secured to the respective side wall component with metal staples, extending through the side wall, tray, and collar, and clinched. The average staple spacing shall be not more than 2-1/2 inches, center to center, with no single spacing exceeding 3 inches. Staples shall have a commercially applied coating or plating such as galvanized (zinc coated) or copper wash. Alternatively, the assembly may be secured to the side wall component by sewing with 15 ounce polished cotton thread. Spacing of stitching shall not exceed 3/8 inch.

3.3.6 Type I, grade E. Drums shall be tight head or removable cover construction as specified (see 6.2). The outer container shall have side walls and headings conforming to the requirements of tables VI, VII, and VIII for the applicable class and size and capacity as specified (see 6.2). The side walls shall be either single or double convolutely wound with no single ply less than 0.012 inch thick and the outer ply waterproofed with all plies firmly glued together. The outer container heads shall be either steel,

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fiberboard or a combination of steel and fiberboard as specified in table VI, VII, and VIII. For class 3 and 4 outer container, the inner surface of steel top heads or covers that may contact the product being packed shall be coated or covered with a flexible sealing disc. The bottom heads shall be as specified in 3.3.6.1 for class 3. The inner linings (class 3) liners or inserts (class 4) and inserts (class 5) shall be as specified in 3.3.6.3 through 3.3.6.5. Closure or method of fastening of heads and covers shall be as specified in 3.3.6.2.

3.3.6.1 Bottom heads and side walls for class 3. Bottom headings shall be made of fiberboard, or a combination of fiberboard and steel, as specified in table VI. The interior side wall and bottom shall be provided with an interior liner or barrier so as to form an integral part of the drum to prevent liquid penetration. The juncture of the bottom heading and side wall shall be so joined as to provide a leakproof seal to the lading. The discs used in the bottom heading shall be constructed of multiple plies of kraft or jute or laminated solid fiberboard, having a combined thickness as specified in table VI. No ply of kraft or jute shall be less than 0.012 inch thick and the plies shall be firmly glued together. The outer ply of the bottom heading shall be water resistant in accordance with commercial standards for fiber drums.

TABLE VI. Side wall and headings for types I, II, and III, grade E, class 3

Maximum content weight (lb) 1/	Maximum capacity (gal)	Minimum side wall bursting strength (psi) 2/	Bottom heading 3/ 4/				
			Fiberboard		Combined		
			Minimum thick (inches)	Minimum bursting strength (psi)	Minimum steel thickness (gage)	Minimum fiberboard bursting strength (psi)	Top cover Steel gage
225	20	1000	0.200	1200	26	800	24
400	35	1100	0.220	1300	24	1000	24
600	55	1200	0.240	1500	24	1000	24

1/, 2/, 3/, 4/ See footnotes to table II.

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TABLE VII. Side wall and headings for types I and II, grade E, class 4

Maximum content weight (lb) 1/	Maximum capacity (gal)	Minimum side wall bursting strength (psi) 2/	Bottom heading 3/ Fiberboard Min thick (in)	Fiberboard Min bursting strength (psi)	Steel Gage	Combination Steel gage	Covers Steel gage
245	15	900	0.200	1200	24	30	24
350	30	1100	0.220	1300	24	30	24
600	55	1200	0.240	1500	24	30	24

1/, 2/, 3/ See footnotes to table II.

TABLE VIII. Side wall and headings for types I and II, grade E, class 5

Maximum content weight (lb) 1/	Maximum capacity (gal)	Minimum side wall bursting strength (psi) 2/	Heading, top and bottom 3/ 6/ Fiberboard Minimum thickness (inches)	Fiberboard Minimum bursting strength (psi)	Steel Min thick (gage)	Combination Min fiberboard thick (in)	Combination Min fiberboard bursting strength (psi)	Combination Min steel thick (gage)
105	6-1/2	600	0.120	800	28	NA 5/	-	-
245	15	700	0.160	1100	26	0.110	400	30 4/
350	30	800	0.160	1100	24	0.110	400	30 4/
450	30	900	0.200	1200	24	0.140	550	30 4/
700	55	1000	0.220	1300	24	0.140	550	30 4/

1/, 2/, 3/, 4/, 5/ See footnotes to table II.

6/ A plastic top cover shall be used for containers with a maximum capacity of 15 gallons. See footnote 4/ to table II for requirements.

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3.3.6.2 Method of fastening top and bottom to side wall. Top and bottom of drums including covers, except class 5, shall be fastened to the side wall so as to meet the requirements of 3.8.1.5 and as specified herein.

3.3.6.2.1 Classes 3 and 4. Covers shall be permanently secured (class 3 only) or removable (class 3 and 4) and shall be fastened by the ring locking method, shall be equipped with a rubber or resilient plastic gasket to effect a liquid tight seal, and shall have either a metal or plastic bung closure, except that lug or friction type fastening may be permitted for drums having a maximum capacity of 20 gallons or less.

3.3.6.2.2 Class 5. Class 5 drums shall be closed in accordance with 3.3.1.3. Removable covers shall be closed in accordance with 3.3.6.2.1 except that the gasket may be omitted.

3.3.6.3 Innermost ply for class 3. The innermost ply of the drum shall be a liquid proof lining constructed of a lamination of plastic film to fiberboard, wound or made as an integral part of the drum. The plastic film shall not affect or be affected by the product being packed.

3.3.6.4 Plastic liner for class 4. The drum shall be equipped with a semirigid plastic liner conforming to the drum contour and meeting the requirements of table IX, or semirigid polyethylene molded one piece fully open head insert or liner having a minimum thickness of not less than 0.015 inch. When used with full open head type plastic insert or liner, tops (covers) shall be equipped with a rubber or resilient plastic gasket to effect a liquid-tight seal. The plastic materials shall not affect or be affected by the product being packed.

TABLE IX. Semirigid plastic liners

Locations in liner	Minimum thickness (inches)	
	Not over 35 gallons	Over 35 gallons but not over 55 gallons
Top chime	0.012	0.015
Side wall measured 2 inches from top	0.015	0.015
Side wall measured from the bottom 2-1/2 inches for under 15 gallons 3-1/2 inches for under 35 gallons and 4 inches for over 35 gallons	0.015	0.015
Side wall measured midway	0.007	0.007

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TABLE IX. Semirigid plastic liners - Continued

Locations in liner	Minimum thickness (inches)	
	Not over 35 gallons	Over 35 gallons but not over 55 gallons
Bottom measured at the curved area of the corner	0.030	0.040
Bottom measured from the side wall 2-1/2 inches for under 15 gallons, 3-1/2 inches for under 35 gallon and 4 inches for over 35 gallons	0.018	0.025

3.3.6.5 Plastic insert for class 5. The fiberboard drum shall be equipped with a molded or thermo formed semirigid one piece plastic liquid tight insert. Openings in the plastic insert shall be no larger than 2.3 inches. Minimum thickness in any part of the insert shall be as specified in table X.

TABLE X. Minimum thickness of plastic inserts for class 5 drum

Capacity (gallons)	Plastic minimum thickness (inches)
6-1/2	0.010
15	0.015
30	0.030
55	0.040

3.4 Construction of type II (normal overseas type).

3.4.1 Type II, grade A drums for dry or solid materials (see 6.1). Drums shall consist of side walls and headings and shall conform to the requirements of table XI for size and capacity as specified (see 6.2).

3.4.1.1 Side wall construction. Method of fastening top and bottom to side wall and interior protection shall be as specified in 3.3.1.1 through 3.3.1.4.

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TABLE XI. Side wall and heads for type II, grade A

Maximum content weight (lb) 1/	Maximum capacity (gal)	Minimum side wall bursting strength (psi) 2/	Heads, top and bottom 4/				
			Fiberboard 3/				
			Minimum thickness (inches)	Minimum bursting strength (psi)	Minimum steel (gage)	Minimum solid wood thickness (inches)	Minimum plywood 3 or more plies (inches)
60	30	500	0.120 or 0.090	300 or 600	30	1/2	3/11
115	45	600	0.160 or 0.120	400 or 800	28	1/2	3/10
150	55	700	0.160 or 0.120	400 or 800	28	1/2	3/8
225	65	800	0.180 or 0.120	500 or 1000	26 5/	1/2 7/	3/8
300	75	900	0.200 or 0.160	550 or 1100	26 5/	25/32	7/16
400	75	1000	0.240 or 0.200	600 or 1200	24 6/	25/32	7/16
550	75	1100	0.220	1300	24 6/	25/32	7/16

1/, 2/, 3/, 4/, 5/, 6/, 7/ See footnotes to table II.

3.4.2 Type II, grade B drums for semiliquid materials (see 6.1). Drums shall consist of side walls and headings and shall conform to the requirements of table XII for size and capacity as specified (see 6.2).

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3.4.2.1 Side wall construction. Method of fastening top and bottom to side wall and interior protection shall be as specified in 3.3.1.1 through 3.3.1.3.

TABLE XII. Side wall and headings for type II, grade B

Maximum content weight (lb) 1/	Maximum capacity (gal)	Minimum side wall bursting strength (psi) 2/	Heads, top and bottom 4/				
			Fiberboard 3/				
			Minimum thickness (inches)	Minimum bursting strength (psi)	Minimum steel (gage)	Minimum solid wood thickness (inches)	Minimum plywood 3 or more plies (inches)
60	10	600	0.160 or 0.120	400 or 800	28	1/2	3/10
115	20	700	0.180 or 0.120	500 or 1000	28	1/2	3/8
150	30	800	0.180 or 0.120	500 or 1000	26 5/	1/2	3/8
225	40	900	0.200 or 0.160	550 or 1100	26 5/	25/32	7/16
300	50	1000	0.240 or 0.200	600 or 1200	24 5/	25/32	7/16
400	60	1100	0.240 or 0.200	900 or 1200	24 6/	25/32	1/2
550	75	1200	0.220	1300	24 6/	25/32	1/2

1/, 2/, 3/, 4/, 5/, 6/ See footnotes to table II.

3.4.3 Type II, grade C drums for hot-poured materials that solidify on cooling (see 6.1.4). Unless otherwise specified, drums shall consist of side walls and headings and shall conform to the requirements shown in table XIII for size and capacity as specified (see 6.2).

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TABLE XIII. Sidewall, fiberboard heading and metal heading for type II, grade C

Maximum content weight (lb) 1/	Maximum capacity (gal)	Maximum inside diameter (in.)	Melting point of packed material (°F)	Sidewall 2/ 3/		Fiberboard, top and bottom 2/ 3/		Metal, top and bottom		Fiberboard, flap top 3/	
				Min thick (in.)	Min dry bursting strength (psi)	Min thick (in.)	Min dry bursting strength (psi)	Min top thickness (gage)	Min bottom thickness (gage)	Min thick (in.)	Min dry bursting strength (psi)
115	-	15	200	0.024	200	0.024	200	33	33	0.024	200
115	-	15	180	0.030	270	0.030	270	33	33	0.030	270
115	-	15	160	0.039	340	0.039	340	32	32	0.039	340
115	-	15	140	0.060	370	0.060	370	28	28	-	-
115	-	15	120	0.090	400	0.090	400	28	28	-	-
-	55	23	160	0.120	750	0.120	900	28	26	-	-
-	55	23	120	0.120	1000	0.120	900	28	24	-	-

1/ The minimum requirements in the above table for side wall, top and bottom are governed by either the weight of contents in first column or by the capacity in second column, whichever requirements are higher.

2/ Cady or Mullen testing method for fiber components (see 4.4.3): Either of the test methods may be used. When more than single ply, test shall be determined from the summation of the tests of individual plies; when test is made on a completed drum, the punctures shall be made from the exterior to the interior surface, in which case the values for side wall shall be not less than 80 percent of the value in the above table. The average of the number of unit findings specified in table XX shall be not less than the prescribed minimum requirements.

3/ For weights of contents of 115 pounds maximum and contents having a melting point of 160°F, the side walls, top and bottom may be 0.045 inch thick with a dry bursting strength of 200 pounds (see 3.3.4).

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3.4.3.1 Side wall construction. Method of fastening top and bottom to side wall and interior protection shall be as specified in 3.3.4.1 through 3.3.4.3.

3.4.4 Type II, grade D drums for rolled or cylindrical items (see 3.3.5, 3.4.2, and 6.1.4). Drums shall consist of side walls and headings (top and bottom) and shall conform to the requirements of table V for size and capacity as specified (see 6.2).

3.4.5 Type II, grade E drums for liquids or articles in liquids (nonregulated) (see 6.1.1). Drums shall consist of side walls and headings (top and bottom). The side wall and headings for class 3 shall conform to the requirements of table VI, for class 4 shall conform to the requirements of table VII, and for class 5 shall conform to the requirements of table VIII for size and capacity as specified (see 6.2).

3.4.5.1 Side wall construction. Method of fastening top and bottom to side wall, covers or top heading, interior protection liners or inserts and bottom heading shall be as specified in 3.3.6 through 3.3.6.5.

3.5 Type III weather resistant overseas type (see 6.1.3 and 6.1.4).

3.5.1 Side wall and headings.

3.5.1.1 Side walls. All component plies of fiberboard as provided in 3.2.1.1 used in side wall shall be combined with a thermoplastic adhesive that will not permit a ply edge separation greater than 1/4 inch when tested as specified in 4.4.3.

3.5.1.2 Top and bottom heading. When fiberboard is used for drum top or bottom, the drum heading shall be laminated with a water resistant adhesive that will not have a ply edge separation greater than 1/4 inch when tested as specified in 4.4.3. The bottom shall consist of multiples of fiberboard with the outer element having a barrier which may be aluminum foil, asphalt, or polyethylene. The outer ply of the bottom shall be scuff and water resistant in accordance with commercial standards.

3.5.1.3 Covers. The drums shall have a full removable cover made from hot dipped galvanized steel of proper gage, fiberboard, or plastic cover as specified for type III, grade A, class 1 drums.

3.5.1.3.1 Cover gasket. The cover shall be fitted with either a round or tubular sponge rubber gasket cemented in place or a cured flowed-in gasket or commercial rubber or other suitable material.

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3.5.1.3.2 Locking band. The cover shall be fastened to the drum by means of a lever actuated locking band. The locking band shall be formed from hot dipped galvanized steel having a minimum thickness before galvanizing of 0.022 inch. The handle and lock shall be formed from steel having a minimum thickness of 0.0533 inch and shall be galvanized. The clamping ring safety latch shall be so constructed that a wire or lead seal may be affixed after closure in such a manner that the cover cannot be removed without destroying the seal.

3.5.1.3.3 Chimes. The top chime used for fastening the cover and locking ring to the side wall and the bottom chime used for fastening bottom to side wall shall be made of hot dipped galvanized steel having a minimum thickness before galvanizing of 0.020 inch. Chime bands shall be made from flat or preformed stock, joined together by welding. The weld may be butt or lap at the option of the contractor. The chime band shall be die crimped or spun into position. The width of the chimes shall be as specified (see 6.2).

3.5.2 Type III, grade A drums for dry or solid materials (see 6.1.3).

3.5.2.1 Class 1, regular construction. Drums shall consist of side walls and headings and shall conform to the requirements of table XIV for size and capacity as specified (see 6.2). Unless otherwise specified (see 6.2), drums shall be provided with a full open top. Barrier or lining shall be as specified in 3.3.1.4.

3.5.2.2 Class 2 foil laminated construction. Class 2 drums shall be as specified for type III, grade A, class 1 (see 3.5.2.1, except that for sidewall, bottom, and cover see 3.5.2.2.1 through 3.5.2.2.3).

3.5.2.2.1 Side wall construction. The side wall shall be a single convolutely wound unit using thermoplastic adhesive or equivalent with no single ply less than 0.012 inch. The construction of the side wall shall consist of the following plies, listed from the inside to the outside of the drum:

a. One ply of 0.002 inch polyethylene coated on 0.00035 inch thick aluminum foil, laminated to 126 pound basis weight kraft sheets with 15 pounds per ream of polymerized petroleum oil fortified with butyl rubber, or 7.2 pounds per ream of polyethylene. Vertical seam to be taped using heat sealed or pressure sensitive tape.

b. Multiple plies of kraft liner board for remainder of sidewall. The minimum side wall test shall meet the requirements of table XIII for the weight of contents specified.

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TABLE XIV. Side wall and headings type III, grade A, class 1 and type III, grade B

Maximum content weight (lb) 1/	Maximum capacity (gal)	Maximum inside diameter (inches)	Fiberboard side wall and headings, top and bottom minimum bursting strength (psi) 2/ 3/		Other headings	
			Average dry	Wet (after 24 hr. immersion)	Min steel thickness (gage)	Plastic cover
115	30	20	800	640	26	4/
150	45	23	900	720	24	4/
225	55	23	1000	800	24	4/
300	65	23	1100	880	24	4/
400	75	23	1200	960	24	4/
550	75	23	1300	1040	24	4/

1/, 2/, 3/, 4/ See footnotes to table II.

3.5.2.2.1.1 Weather resistant containers. When specified (see 6.1 and 6.2), side wall construction shall be as specified for 550 pounds, except that when gallonage capacity does not exceed 55 gallons, interior side wall and bottom of drum shall be laminated with aluminum foil to provide a ground from the interior surface of the drum to the outer surface of the bottom chime. The net weight of contents shall not exceed 600 pounds.

3.5.2.2.2 Bottom. The drum bottom shall be as specified in 3.5.1.2 except that the bottom shall include on the inside surface a polyethylene/aluminum/kraft liner disc constructed as specified in 3.5.2.2.1(a) above and meet the Mullen requirements of table XIV for the maximum net weight of contents specified.

3.5.2.2.3 Cover. The drums shall have a full removable cover made from hot dipped galvanized steel, complying with gage requirements of table XIV.

3.5.2.2.3.1 Cover gasket (see 3.5.1.3.1).

3.5.2.2.3.2 Locking band (see 3.5.1.3.2).

3.5.2.2.3.3 Chimes (see 3.5.1.3.3).

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3.5.2.2.4 Size and content. Except as specified in 3.5.2.2.1.1, the size of the drums shall be as specified by the procuring activity (see 6.2). The net weight of contents shall be maximum of 550 pounds.

3.5.3 Type III, grade B drums for semiliquid materials). Drums shall consist of side walls and headings and shall conform to the requirements of table XIV for size and capacity as specified (see 6.2).

3.5.3.1 Side wall, top, and bottom. Side wall construction and method of fastening top and bottom to side wall shall be as specified in 3.5.1 through 3.5.1.3.3.

3.5.3.2 Barrier, lining, or interior coating. The drum must be so treated, processed or proofed, or equipped with a plastic film bag type liner as to prevent absorption of contents.

3.5.4 Type III, grade E drums for liquids or articles in liquids (nonregulated). Drums shall be tight head or removable cover construction as specified (see 6.2). The outer container shall have side walls and headings conforming to the requirements of table VI for class 3, table XV for class 4, and table XVI for class 5 for the applicable class and size and capacity as specified (see 6.2). The side walls shall be single convolutely wound with no single ply less than 0.012 inch thick. All plies shall be firmly glued together (see 3.5.1.1), and when specified the outer ply shall be water resistant in accordance with the following commercial standards for fiber drums: Rosin sizing, or high alkaline resistant sizing, wax coating, or the application of clear lacquer to the outer ply will be acceptable. The coating used shall not increase the flammability of the drum. The outer container heads shall be either steel, fiberboard, combination of steel and fiberboard or plastic as specified in table VI, XV, and XVI. For class 3 and 4 outer container, the inner surface of steel top heads or covers that may contact the product being packed shall be coated or covered with a flexible sealing disc. The bottom heads shall be as specified in 3.5.4.1 for class 3. The inner linings (class 3) liners or inserts (class 4) and inserts (class 5) shall be as specified in 3.5.4.4 through 3.5.4.6. Closure or method of fastening of heads and covers shall be as specified in 3.5.4.2.

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TABLE XV. Side wall and headings for type III, grade E, class 4

Maximum content weight (lb) 1/	Maximum capacity (gal)	Minimum side wall bursting strength (psi) 2/	Bottom headings 3/ Fiberboard Min thick (in)	Fiberboard Min bursting strength (psi)	Steel gage	Combination Fiber thick (in)	Combination Fiber burst strength (psi)	Steel gage	Covers 4/ Steel gage
245	15	900	0.200	1200	24	0.110	440	30	24
350	30	1100	0.220	1300	24	0.140	550	30	24
600	55	1200	0.240	1500	24	0.160	900	30	24

1/, 2/, 3/, 4/ See footnotes to table II.

TABLE XVI. Side wall and headings for type III, grade E, class 5

Maximum content weight (lb) 1/	Maximum capacity (gal)	Minimum side wall bursting strength (psi) 2/	Heading, top and bottom 3/ 4/ Fiberboard Minimum thickness (inches)	Fiberboard Minimum bursting strength (psi)	Steel Min thick (gage)	Combination Min fiberboard thick (in)	Combination Min fiberboard bursting strength (psi)	Min steel thick (gage) 5/
245	15	700	0.160	1100	26	0.110	400	30
350	30	900	0.160	1100	24	0.110	550	30
450	30	1000	0.200	1200	24	0.140	550	30
700	55	1200	0.220	1300	24	0.140	550	30

1/, 2/, 3/, 4/ See footnotes to table II.

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5/ For combinations, fiberboard and steel bottoms, the steel shall be locked into the crimp. Flat steel discs are not acceptable.

3.5.4.1 Bottom heads and side walls for class 3. Bottom headings shall be made of fiberboard, or a combination of fiberboard and steel, as specified in table VI. The innermost ply of the bottom heading shall be a liquid proof lining of plastic film. The interior side wall and bottom shall be provided with an interior liner or barrier so as to form an integral part of the drum to prevent liquid penetration. The juncture of bottom heading and side wall shall be so joined as to provide a leakproof seal to the lading. The discs used in the bottom heading shall be so constructed of multiple plies of kraft or laminated solid fiberboard, having a combined thickness as specified in table VI and conforming to 3.5.1.3. Side walls shall be combined as specified in 3.5.1.2 using thermoplastic adhesive.

3.5.4.2 Method of fastening top and bottom to side wall. Top and bottom of drums including covers, except class 5, shall be fastened to the side wall so as to meet the requirements of 3.8.1.7 and as specified herein.

3.5.4.2.1 Classes 3 and 4. Covers shall be as specified in 3.3.6.2.1 except they shall be made from hot dipped galvanized steel of proper gage or plastic cover as specified in table VI for type III, class 3 and table XV for type III, class 4 drums.

3.5.4.2.2 Class 5. Class 5 drums shall be closed in accordance with 3.2.1.3. Removable covers shall be closed in accordance with 3.3.6.2.2.

3.5.4.3 Chimes for classes 3, 4, and 5. Classes 3, 4, and 5 drums shall be constructed with metal top and bottom chimes as shown in style 1, figure 3.

3.5.4.4 Inner ply for class 3. The innermost ply shall be constructed as specified in 3.3.6.3.

3.5.4.5 Plastic liner for class 4. The drum shall be equipped with a semirigid plastic liner as specified in 3.3.6.4.

3.5.4.6 Plastic insert for class 5. The fiberboard drum shall be equipped with a molded or thermo formed rigid one piece plastic closed head insert as specified in 3.3.6.5.

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3.6 Closures.

3.6.1 Friction cap type closures.

3.6.1.1 Types I and II, grades A, B, and C drums, and type III, grades A and B. When specified (see 6.2), friction cap type closures may be used in lieu of top headings and closures specified herein, provided the friction cap is of metal not less than 28 gage in thickness. All friction caps shall be securely held in place by crimped or bolted type lugs, or by use of a plug expanding tool, in such a manner as to withstand the tests specified in section 4.

3.6.1.2 Types I, II and III, grade E, classes 3 and 4 drums. Lug or friction type fastening may be used in lieu of top closures specified herein for drums having a maximum capacity of 20 gallons or less. The friction caps shall be applied in such a manner as to withstand the applicable drop test specified in 4.4.3.

3.6.2 Types I and II, grade D drums. When this specification is referenced in documents for procurement or shipment of an item, figure 1 shall apply for closure of the packed drum.

3.6.3 Type III drums. When this specification is referenced in documents for procurement or shipment of an item, style 1, figure 3 shall apply for closure of the packed drum.

3.7 DOT specifications (all types). When required by the nature of the contents, drums shall also comply with the minimum requirements of the Department of Transportation. Where a conflict in minimum requirements exists between the DOT specifications (49 CFR 178.224 and 178.225) and this specification, title 49 CFR shall govern.

3.8 Performance.

3.8.1 Drop strength (see 4.5.4).

3.8.1.1 Type I, grade A, class 1 and grade B (domestic type). Grade A, class 1 and grade B drums shall withstand the drop test specified in 4.4.3 without leakage of contents.

3.8.1.2 Type I, grade A, class 2 (domestic type) and type III, grade A, class 2. Type I, and type III, grade A, class 2 drums shall withstand the drop test specified in 4.4.3 without sifting or leakage of contents. The same drum shall not be used for both the drop test and the water vapor transmission rate test (see 4.5.6).

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3.8.1.3 Type I, grade D (domestic type). Grade D drums shall withstand the drop test specified in 4.4.3 without shearing of side wall, dislocation of heading disc, splitting of heading or disc, or separation of component drum parts. Shearing of metal staples or cotton thread stitching at points of impact shall not be considered as a separation of component parts (see 4.5.4.2).

3.8.1.4 Type II, grades A, B, and D (normal overseas type). Grades A, B, and D drums shall withstand the drop test specified in 4.4.3 without leakage of contents.

3.8.1.5 Type I, grade E (domestic type) and type II, grade E (normal overseas type). The drums shall withstand the drop test specified in 4.4.3 without leakage of contents.

3.8.1.6 Type III, grades A and B drums (weather resistant type).

3.8.1.6.1 Drop strength, dry. Drums shall withstand the drop test (dry) specified in 4.4.3, without leakage of contents or separation of component parts, or entrance of water to interior surfaces of the drum.

3.8.1.6.2 Drop strength, wet. Drums shall withstand the drop test (wet) specified in 4.4.3, without leakage of contents or separation of component parts.

3.8.1.7 Type III, grade E drums (weather resistant type). The drums shall withstand the drop test specified in 4.4.3.

3.8.2 Compression.

3.8.2.1 Type I, grade C (domestic type) and type II, grade C (normal overseas type). Grade C drums, containing products having a minimum melting point of 120°F shall withstand the compression test specified in 4.4.3, without bursting the side wall or causing appreciable change in appearance of the drum.

3.8.2.2 Type III, grades A, B and E (weather resistant). All grade drums shall withstand the dry and wet compression tests specified in 4.4.3 as applicable, without buckling of the side walls sufficient to cause damage to their contents.

3.8.3 Water vapor transmission rate (WVTR).

3.8.3.1 Type I, grade A, class 2 drums (domestic type) and type III, grade A, class 2 drums. The water vapor transmission rate, of the drums shall not exceed 0.075 gram per 100 square inches per 24 hours when tested as specified in 4.4.3.

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3.9 Drum capacity.

3.9.1 Types I and II, grade B drums. When drums are to be used for contents purchased by the gallon, the actual capacity shall be not less than rated capacity plus outage or head space specified in table XVII.

3.9.2 Types I, II, and III grade E drums. When drums are to be used for liquids, the actual capacity shall be not less than rated capacity plus outage or head space specified in table XVII.

TABLE XVII. Drum capacity for types I and II, grades B and E drums and type III, grade E drums

Drum capacity, net (gallons)	Outage allowance 1/	
	Minimum (inches)	Maximum (inches)
Up to 10	1/2	1-1/2
Over 10 up to 30	1	2
Over 30	1-1/2	2-1/2

1/ The indicated requirements are for dry or solidified products. For wet materials the outage shall be not less than 4 percent nor more than 5 percent of the rated capacity.

3.10 Drum diameter, types I, II and III. Unless otherwise specified, drum diameter shall be as specified in the contract, order, or specification for the commodity being packed, with a tolerance of minus 1/16 inch and plus 1/4 inch (see 6.2).

3.11 Drum height, types I, II and III. The inside height of the drum shall be as specified in the contract, order, or specification for the commodity being packed, with a tolerance of no minus and plus 3/8 inch (see 6.2).

3.12 Compliance marking. In addition, and adjacent to the drum manufacturer's marking required by ASTM D 3951 or 49 CFR Transportation Part 171-179, each drum shall have stamped or printed on the side in black capital letters not less than 3/16 inch high, the applicable information on figure 4 with the applicable amounts inserted.

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3.12.1 Precautionary marking (grade D only). In addition, the following marking shall be stenciled or printed on the cover side wall of each fiberboard drum in letters not less than 3/4 inch in height:

STAND ON END

KEEP COOL AND DRY

3.13 Workmanship. The fiber drums shall conform to the quality of product established by this document and the occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirement in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for assuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point or at all points in the manufacturing process necessary to assure compliance with all dimensional requirements.

4.1.3 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

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4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in 4.4.2 and shall be tested for the characteristics specified in 4.4.3. The presence of any defect or failure of any test shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.4.1.1 Examination of components and materials. In addition, components and materials shall be examined as specified in table XVIII. Failure to meet any requirement shall be cause for rejection of the component or material lot.

4.4.1.2 Reclaimed fiber component. The bidder shall submit a certificate of compliance indicating that the material offered to the Government contains a minimum of 5 percent reclaimed fibers as specified in 3.2.2.

TABLE XVIII. Examination of component or material

Examine 1/	Sample	Requirement paragraph
Low carbon steel:		
Thickness	1 sheet (each gage)	3.2.1
Material		
Kraft liner board:		
Weight	1 sheet	3.3.2.1
Aluminum barrier board:		
Thickness	1 sheet	3.3.2.1
Material		
Cover gasket:		
Size	5 gaskets	3.3.2.3.1
Material		
Thread:		
Weight	1 spool	3.3.5.2.5

1/ Material identification shall be made by checking invoices and bills of lading.

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4.4.2 End item visual examination. The end item shall be examined for the defects listed in table XIX. The lot size shall be expressed in units of drums. The sample unit shall be one drum. The inspection level shall be S-3 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 4.0 for major defects and 6.5 for total (major and minor combined) defects.

TABLE XIX. Examination of end item

Examine	Defect	Category	
		Major	Minor
Side walls and cover side wall (as applicable)	Plies of each unit not firmly glued throughout entire area	101	
	Plies not as specified	102	
	Overlap of plies less than 4 inches		201
	Barrier, lining, or interior coating missing when specified	103	
	Barrier, lining, or interior coating not material or construction specified	104	
	Plastic liner (when specified) less than minimum thickness specified	105	
	Opening in plastic component more than 2.3 inches		202
	Side wall thickness less than specified		203
	Plastic component missing when applicable	106	
Type I, grade D only	Cover side wall less than 4-3/4 inches in inside depth	107	
Overlap joint (as applicable)	Overlap less than 1 inch	108	
	Not firmly glued throughout entire area of contact	109	
Type I, class 2 only:			
Bottom	Plies not firmly glued throughout entire area	110	
	Plies not laminated in sequence specified	111	
	Total thickness less than 0.160 inch	112	

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TABLE XIX. Examination of end item - Continued

Examine	Defect	Category	
		Major	Minor
Type I, class 2 only - continued			
Cover	Not fully removable when specified	113	
	Not 26 gage steel	114	
	Indication of grease		204
	Finish coating or sealing disc, as applicable, missing or not applied over complete surface		205
Cover gasket	Not type specified	115	
Locking band	Not lever actuated type	116	
	Components less than specified thickness of steel	117	
	Finish missing or not applied over complete surface		206
	No means for affixing wire and seal as required		207
Chime	Less than specified thickness	118	
	Weld not continuous	119	
	Seam other than butt or lap		208
	Crimp incomplete (not joining parts securely)	120	
Types I and II, grade D only			
Heading (discs)	Combined thickness of plies less than specified	121	
	Adhesive between plies not water resistant when specified	122	
	Outer ply not water resistant	123	
	Joints of wood discs not tightly glued (water resistant glue when specified)	124	
Tray	Fiberboard less than 0.025 inch thick	125	
	Material not kraft		209

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TABLE XIX. Examination of end item - Continued

Examine	Defect	Category	
		Major	Minor
Types I and II, grade D only - continued			
Collar	Not as specified (see 3.3.5.2.4)		210
	Thickness less than specified	126	
	Width less than 5/8 inch or more than 1 inch	127	
Closures	Not type specified	128	
Friction cap	Metal less than 28 gage	129	
	Method for holding cap not as specified		211
Assembly	Not assembled in sequence specified	130	
	Staples when specified more than 3 inches apart		212
	Stitches when specified more than 3/8 inch apart		213
	Not liquid tight when specified	131	
Closure (as applicable) (when furnished as component)	Not two continuous wraps of tape specified	132	
	End point of tape does not overlap starting point by at least 2 inches		214
	Tape less than 3 inches wide	133	

4.4.3 End item testing. The finished drum shall be tested as specified in table XX. Five sample drums randomly selected throughout the lot shall form the sample size. Any nonconformance shall result in rejection of the lot.

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TABLE XX. Testing of the end item

Characteristic	Requirement paragraph	Test method	No. of determinations	Results reported as: Pass or fail	Results reported as: Numerically to nearest
All types, grades, and classes, as applicable, side wall and headings:					
Bursting strength of fiberboard:					
Dry	Tables II thru XVI as applicable	4.5.1	Average of 10 1/		Whole psi
Wet	Table XIV	4.5.1	Average of 10 1/		Whole psi
Puncture strength of fiberboard:					
Dry	Table V, as applicable	4.5.2	Average of 5 1/		Unit
Drop test:					
Types I and II	3.8.1.1 3.8.1.2 3.8.1.3 3.8.1.4 3.8.1.5	4.5.4.1 4.5.4.7 4.5.4.2 4.5.4.2 4.5.4.6	2 drops	X	
Type III					
Dry	3.8.1.6.1 3.8.1.7	4.5.4.3 4.5.4.6.1	2 drops	X	
Wet	3.8.1.6.2	4.5.4.4	2 drops	X	
Compression test:					
Dry	3.8.2.1	4.5.5.1	1	X	
	3.8.2.2	4.5.5.2.1	1	X	
Wet	3.8.2.2	4.5.5.2.2	1	X	

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TABLE XX. Testing of the end item - Continued

Characteristic	Requirement paragraph	Test method	No. of determinations	Results reported as: Pass or fail	Results reported as: Numerically to nearest
Types I and III, grade A, class 2:					
Water vapor transmission rate	3.8.3.1	4.5.6	1	X	
Type III only:					
Ply separation and bonding	3.5.1.1 3.5.1.2	4.5.3	1	X	

1/ Determinations shall be made on specimens from a single drum.

4.4.4 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Packing	Cover or heading missing when specified Packing not as specified
Marking	Omitted, incomplete, incorrect, illegible, of improper size, location, sequence, or method of application

4.5 Methods of inspection.

4.5.1 Bursting strength.

4.5.1.1 Apparatus (Mullen or Cady type). Bursting strength of the fiberboard shall be determined in accordance with Method 2007 of FED-STD-101. To comply with the requirements of this specification, the average of the 10 bursts made from the exterior to the interior of the drum specimens shall

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be not less than the strength requirements in tables II through XVI, (3.3.1 through 3.5.4) inclusive (as applicable) for the specified type, grade, class, and capacity drum.

4.5.2 Puncture resistance test.

4.5.2.1 Apparatus (Beach tester). The puncture resistance of the fiberboard shall be in accordance with TAPPI Standard T 803 M 50. To comply with the requirements of this specification, the average of the tests made from the exterior to the interior of the drum specimens shall be not less than the strength requirements in table V for the specified type, grade, class, and capacity drum.

4.5.3 Ply separation and bonding test. Specimens of the fiberboard shall be tested for ply separation and satisfactory bonding in accordance with Method 3028, procedure A of FED-STD-101. The submersion cycle shall be not less than 24 hours. Separation of the plies shall extend not more than 1/4 inch from the edge of the specimen when edges are flexed by pressure from the thumb (see 3.5.1.1). All other areas shall remain bonded.

4.5.4 Drop test. Drop tests shall be in accordance with Method 5007 of FED-STD-101 except as specified herein. Drums shall be filled to the maximum authorized weight with the designated contents (see 6.2). If procurement document does not indicate what contents will be, or if procurement is for stock purposes and more than one content may eventually be used, the following contents shall be used:

Grade A drums - a mixture of sand, sawdust, and flour, or equivalent material, shall be mixed to arrive at necessary weight or capacity.

Grade B drums - asphalt or lubricating grease.

Grade D drums - when product to be packed is unknown to the drum supplier, drums shall be filled with a rolled material equal, in length to the inside length of the drum, with an allowable tolerance of 1/4 inch and equal in diameter to the inside diameter of the drum, with an allowable tolerance of 1/4 inch. The weight of the rolled material shall be as follows:

Inside diameter of drum up to and including 11 inches - 100 pounds, tolerance ± 19 percent.

Inside diameter of drum over 11 up to and including 14 inches - 150 pounds, tolerance ± 10 percent.

Inside diameter of drum over 14 up to and including 16 inches - 250 pounds, tolerance ± 10 percent.

When the specified product is known, the drums shall be filled with the specified product of a dummy load equivalent in weight and conformation. Type I grade D drums shall be closed in accordance with figure 1 of this specification. Closure of other drums shall be as specified for the applicable drum. Drums filled to the maximum authorized weight with its

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designated contents, or equivalent material, and closed, as specified in 3.3.1.3, shall be placed on a top end edge and dropped on a solid concrete floor or steel plate of sufficient mass to absorb shock without deflection, so as to strike diagonally on the top closure or edge of the chime. On lapped joint side walls, the impact shall be adjacent to the lapped joint. The drums shall then be balanced and dropped from the same height on a bottom end and edge diagonally opposite from the first edge.

4.5.4.1 Type I, grade A, class 1 and grade B (domestic type). The height of drop shall be sufficient to provide not less than 160 foot pounds impact. The height of drop in inches shall be equivalent to 1920 divided by the weight of the drum and contents in pounds, except that the minimum height of the drop shall be 12 inches and the maximum height shall be 24 inches (see 3.8.1.1).

4.5.4.2 Type I, grade D and type II, grades A, B, and D. The height of the drop shall be sufficient to provide not less than 320 foot pounds impact. The height of the drop in inches shall be equivalent to 3840 divided by the weight of the drum and contents in pounds, except that the minimum height of the drop shall be 18 inches and the maximum height shall be 36 inches.

4.5.4.3 Type III, grades A and B (dry). The height of the drop in inches shall be sufficient to provide not less than 400 foot pounds impact. The height of the drop shall be equivalent to 4800 divided by the weight of the drum and contents in pounds, except that the minimum height of the drop shall be 24 inches and the maximum height shall be 48 inches.

4.5.4.4 Type III, grades A and B (wet). The packed drum shall be subjected to a continuous water spray for 120 hours. The spray nozzles shall be located not less than 6 feet above the containers and designed to break up the water into a heavy mist of very small droplets. The nozzle design shall be such that no large drops of water impinge on the test containers at a high velocity and such as to allow the drops to fall vertically. The nozzles shall be of such size, and so spread, that 2.5 gallons $\pm$ 0.5 gallon of water per hour fall uniformly distributed on each square foot of floor area. The water used in the spray system shall be maintained at a constant temperature of 73 $\pm$ 2^{O}F. Containers shall be placed in the shower chamber in the normal stacking position and so arranged that the distance between adjacent containers is not less than 6 inches. On removal from the shower chamber, the drum shall be dropped on a solid concrete floor or steel plate, of sufficient mass to absorb shock without deflection so as to stike diagonally on the top edge from a height of not less than 24 inches and not more than 48 inches and not to exceed 400 foot pounds impact. The same test shall be made on the bottom diagonally opposite from the first drop. In order to test for entrance of water into the drum, type III, grades A and B drums shall again be placed in a continuous 4 hour shower test and then be inspected for entrance of water on the interior surface of the drum (see 3.8.1.6.2).

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4.5.4.5 Drums with wood headings. Drums with wood headings shall have the cover attached so that the grain direction of the wood in the cover and in the bottom heading shall be parallel with each other. Drums with wood headings shall be dropped with the grain of the wood in the cover parallel to the concrete surface on which dropped (see figure 2).

4.5.4.6 Types I and II, grade E. Drums filled to net capacity with water must withstand without leakage a tip-over fall on the cover chime on solid concrete followed by a diagonal drop on the bottom chime sufficient to provide an impact of not less than 500 foot pounds, except that the maximum height of drop shall not exceed 2 feet (see 3.8.1.5).

4.5.4.6.1 Type III, grade E. Drums filled to net capacity with water must withstand without leakage a tip-over fall on the cover chime on solid concrete followed by a diagonal drop on the bottom chime sufficient to provide an impact of not less than 500 foot pounds, except that the maximum height of drop shall not exceed two feet for classes 3 and 4. For class 5, the height of drop shall be 48 inches (see 3.8.1.7).

4.5.4.7 Types I and III, grade A, class 2 drums. The drums shall be filled to maximum authorized weight with a dry, finely powdered material. The filled drum shall withstand two drops from a height of not less than 4 feet on a solid concrete floor or steel plate, of sufficient mass to absorb the load without deflection. The first drop shall be made diagonally on the bottom chime, and the second drop shall be made diagonally on the top chime. The drum shall show no evidence of leakage of finely powdered material, delamination, or other physical defects (see 3.8.1.2).

4.5.5 Compression test.

4.5.5.1 Type I, grade C (domestic type) and type II, grade C (normal overseas type). Grade C drums shall be filled with asphalt, wax, or other material, with a melting point of not more than 125°F. A uniformly distributed static load of 350 pounds shall be applied on the top of the 115 pound capacity drum, and 900 pounds on top of the 55 gallon capacity drum, for a period of 24 hours. Prior to application of the load, the filled drum shall be conditioned so that contents and drum shall have a temperature of 100 ± 5°F and the test shall be conducted in a room maintained at a temperature of 100 ± 5°F (see 3.8.2.1).

4.5.5.2 Type III, grades A, B and E drums (weather-resistant overseas type).

4.5.5.2.1 Dry. Drums, empty and properly closed, shall be subjected to a uniformly distributed top compression load gradually applied, either static or dynamic as specified in table XXI. The speed of the gradually applied top shall be regulated to 1/2 ± 1/4 inch per minute. The static load shall remain in place for 24 hours (see 3.8.2.2).

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4.5.5.2.2 Wet. Drums, empty and properly closed, shall be subjected to a continuous water spray shower, as specified in 4.5.4.4. Immediately after removal from the shower chamber, a uniformly distributed load, either static or dynamic as specified in table XXI shall be applied to the top of the drum. The duration of the static load and the speed of application of the dynamic load shall be as stated in 4.5.5.2.1 (see 3.8.2.2).

TABLE XXI. Minimum top load resistance for type III grades A and B; type III, grade E, classes 3, 4 and 5

Maximum net weight (pounds)	Dry		Wet	
	Top load static (lb)	Top load dynamic (lb)	Top load static (lb)	Top load dynamic (lb)
Type III, grades A and B:				
115	1500	2000	750	1500
150	1800	2400	900	1800
225	2100	2800	1050	2100
300	2250	3000	1125	2250
400	2400	3200	1200	2400
550	2475	3300	1240	2475
Type III, grade E:				
Class 3 only 225	2100	2800	1050	2100
Classes 4 and 5 245	1800	2400	900	1800
Classes 4 and 5 350	2250	3000	1125	2250
Class 3 only 400	2250	3000	1125	2250
Class 5 only 450	2250	3000	1125	2250
Classes 3 and 4 600	2400	3200	1200	2400
Class 5 only 700	2400	3200	1200	2400

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4.5.6 Water vapor transmission rate (see 3.8.3).

4.5.6.1 Type I, grade A, class 2 drums (domestic type) and type III, grade A, class 2 drums. The water vapor transmission rate of the drums shall be determined in accordance with ASTM D 1008, using procedure A for reclosable containers. The weighing frequency shall be biweekly. Water vapor transmission rate calculations shall be made for the period of constant rate of gain, as indicated by at least three successive points in a straight line. The slope of this portion of the curve shall furnish a measure of water vapor permeability.

4.5.6.2 Repeated tests. The water vapor transmission rate test need not be repeated for new orders provided the materials and fabrication of the drums have not been changed, and a certified statement to this effect is furnished the preparing activity along with a copy of the report of the tests made. However, the procuring activity shall have the right to require the performance of the water vapor transmission rate test at any time when it is deemed necessary to determine the conformance of the drums to the requirements of this specification.

4.5.7 Melting point of paraffin. The melting point of paraffin shall be determined in accordance with ASTM D 87.

5. PACKAGING

(The requirements specified herein apply only for direct Government procurements).

5.1 Packing. Packing shall be Commercial. Each drum requiring a cover or heading shall have the cover or heading applied or accompany the drum during shipment.

5.1.1 Commercial packing. Drums shall be packed in accordance with ASTM D 3951.

5.2 Palletization. Drums shall be palletized on a 4-way entry pallet in accordance with load type XIIa, XIIb, or XVII of MIL-STD-147. Pallet types shall be type I (4-way entry), type IV, or type V in accordance with MIL-STD-147. Pallets shall be fabricated from wood groups I, II, III, or IV of MIL-STD-731. Load types XIIa and XIIb shall be bonded with primary and secondary straps in accordance with bonding means K and L, or film bonding means O or P.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special markings required by the contract, shipments shall be marked in accordance with FED-STD-123 or ASTM D 3951, as applicable.

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5.3.2 Military agencies. In addition to any special markings required by the contract, shipments shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

6.1 Intended use.

6.1.1 Domestic type (type I). Fiberboard drums covered by this specification are intended for use as domestic shipping containers. Grade A drums are for dry or solid materials. Class 1 regular construction should normally be used; class 2 foil laminated should be used for highly hygroscopic materials needing a barrier, such as desiccants. Use of grade B drums should be limited to semisolid materials having a consistency similar to asphalt, lubricating grease, petrolatum, refractory cement, caulking compounds, roof coatings, adhesives, textile sizing, and food products such as jellies and fondants. The consistency of asphalt materials and lubricating greases should not exceed 350 units when measured in accordance with method 311.6 of FED-STD-791 (ASTM D 217). Class 2 is normally used when high moisture barrier is required or when the type I, grade A, class 2 is supplemental by the aluminum liners it may be used to deter electrostatic charges and prevent explosion. Grade C drums are for hot-poured materials that solidify on cooling and should be limited to materials poured at temperature not exceeding 400 F^{o}. Grade E drums are for nonregulated liquids or articles in liquids.

6.1.1.1 Type I, grade D. This is also intended for normal overseas shipment as described in 6.1.2. When substituted for type II drums, the drums shall be marked type II, overseas type (nonweather resistant).

6.1.2 Type II overseas (nonweather resistant). Fiberboard drums covered by this specification are intended for use for normal overseas shipment where numerous handlings and storage at destination are not anticipated. Usage factors are the same as specified in 6.1.1. They are not to be used for overseas shipment of hazardous material, i.e., ammunition, explosives, propellants, etc. They may, however, be used for domestic shipment of these materials subject to restrictions of 49 CFR 100-199.

6.1.3 Type III overseas (weather resistant). Fiberboard drums covered by this specification are intended for unprotected weather exposure usage for added protection in high humidity or outdoor storage environment. They are not to be used for overseas shipment of hazardous material, such as ammunition, explosives, or propellants. They may, however, be used for domestic shipment of these materials subject to restrictions of 49 CFR 100-199.

6.1.4 Exceptional requirement. Exceptional commodities, especially dangerous articles (hazardous material), may require better material and construction requirements than are covered by this specification. In every

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case, the drums must conform to the specifications prescribed in the Department of Transportation regulations (DOT 49 CFR 100-199) governing the transportation of the particular articles or materials being packed.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

a. Title, number, and date of this specification.
b. Part number, type, grade, and class required (see 1.2.1 and 1.2.2).
c. When a first article is required (see 3.1, 4.3, and 6.4).
d. Percentage of reclaimed fibers required if other than that specified (see 3.2.2).
e. Size and capacity required (see 3.3.1, 3.3.2.4, 3.3.3, 3.3.4, 3.3.5, 3.3.6, 3.4.1, 3.4.2, 3.4.3, 3.4.4, 3.4.5, 3.5.2.1, 3.5.2.2.4, 3.5.3, and 3.5.4).
f. Selection of material for heads (see 3.3.1).
g. The type of interior protection required for the specific use of the drum (see 3.3.1.4).
h. Construction of inner ply (see 3.3.2.1).
i. When type I, grade A, class 2 with aluminum foil liner may be used as a type I, class 2 drum for product requiring a positive ground (see 3.3.2.1.1).
j. When side wall 0.041 inch is required for hot-poured materials (see 3.3.4).
k. For grade C drums, whether top heading is required and whether other construction shall be permitted (see 3.3.4, 3.4.3).
l. When grade D drums are required to have spirally wound side walls (see 3.3.5.1).
m. For grade E drums, whether tight head or removable cover is required (see 3.3.6 and 3.5.4).
n. Width of chimes for type III drums (see 3.5.1.3.3).
o. When other than open top is required (see 3.5.2.1).
p. When friction cap type closures shall be used (see 3.6.1.1).
q. Contents for drop test (see 4.5.4).

6.3 Additional protection. When fiber drums are being procured for packing a specific product that may require additional protection against contamination or deterioration, it should be stipulated that the drum shall be furnished with a barrier, lining, or an interior coating that will provide such protection and which will not adversely affect, or be adversely affected by the contents (see 3.3.1.4 and 3.3.3.2).

6.4 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for selection, inspection, and approval of the first article.

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6.5 Reclaimed fibers identification. Reclaimed fibers shall be derived from one or more of the following sources as applicable:

PART I

(a) Paper, paperboard and fibrous wastes from factories, retail stores, office buildings, homes, etc., after they have passed through their end-usage as a consumer item including:

1. Used corrugated boxes;
2. Old newspapers;
3. Old magazines;
4. Mixed waste paper;
5. Tabulating cards; and
6. Used cordage.

(b) All paper, paperboard, and fibrous wastes that enter and are collected from municipal solid waste.

PART II

(a) Dry paper and paperboard waste generated after completion of the paper-making process* including:

1. Envelope cuttings, bindery trimmings, and other paper and paperboard waste, resulting from printing, cutting, forming, and other converting operations;
2. Bag, box and carton manufacturing waste; and
3. Butt rolls, mill wrappers, and rejected unused stock.

(b) Finished paper and paperboard from obsolete inventories of paper and paperboard manufacturers, merchants, wholesalers, dealers, printers, converters, or others.

(c) Fibrous by-products by harvesting, manufacturing, extractive, or woodcutting processes, flax straw, linters, bagasse, slash, and other forest residues.

(d) Wastes generated by the conversion of goods made from fibrous materials; i.e., waste rope from cordage manufacture, textile mill waste, and cuttings.

(e) Fibers recovered from waste water which otherwise would enter the waste stream.

* The papermaking process is defined as those manufacturing operations up to and including the cutting and trimming of the paper machine reel into small rolls or rough sheets.

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6.6 Subject term (key word) listing.

Drums
Fiberboard containers
Shipping containers

Custodians:

Army - GL
Navy - SA
Air Force - 69

Review activities:

Army - MD, SM, EA
DLA - GS
Air Force - 43, 70, 71, 80, 82, 84, 99

User activities:

Navy - OS, SH

Preparing activity:

Army - GL

Civil Agency Coordinating Activity:

GSA - FSS

Project No. 8110-0288

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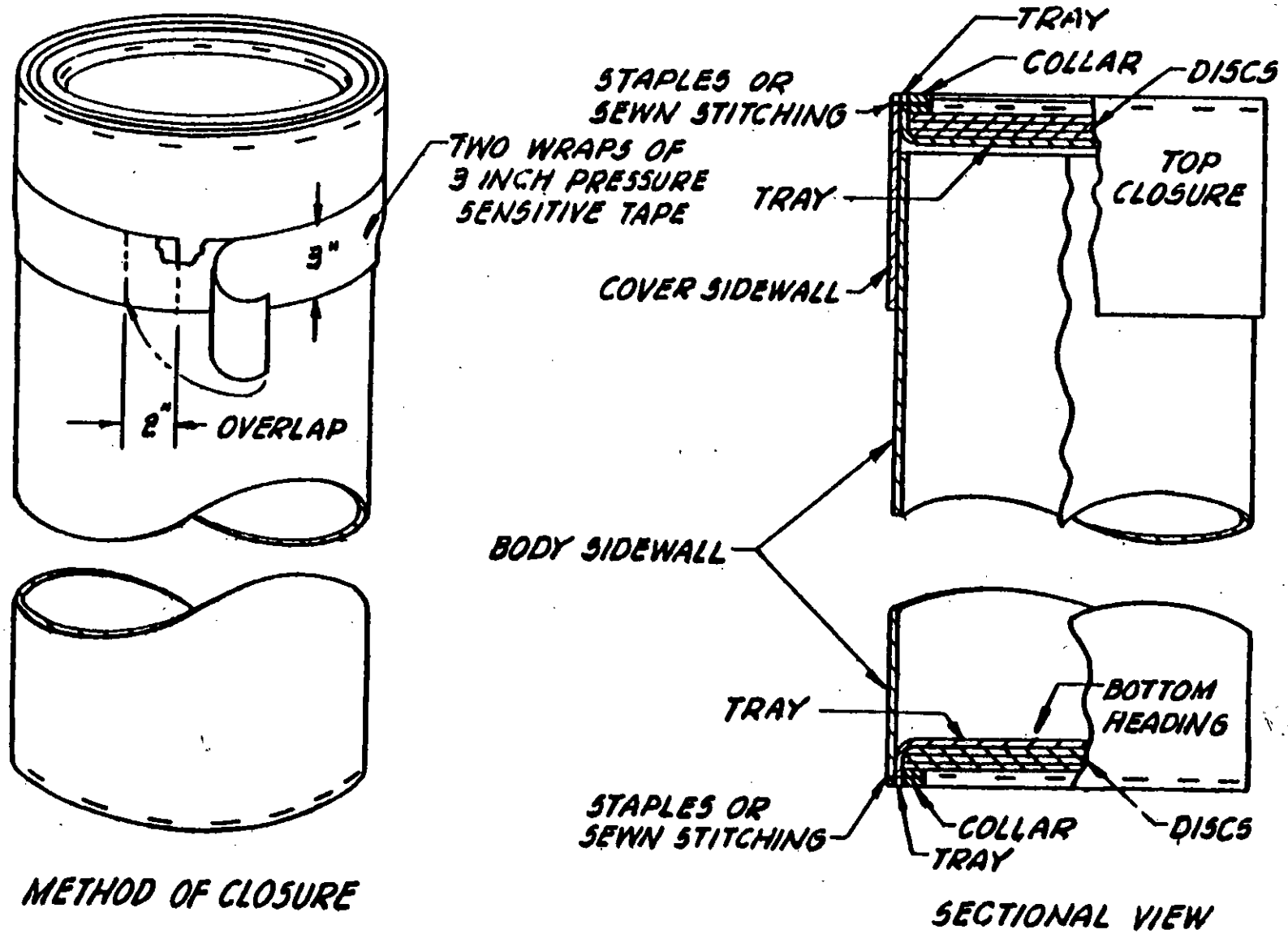


FIGURE 1. Drums, fiber, grade D.

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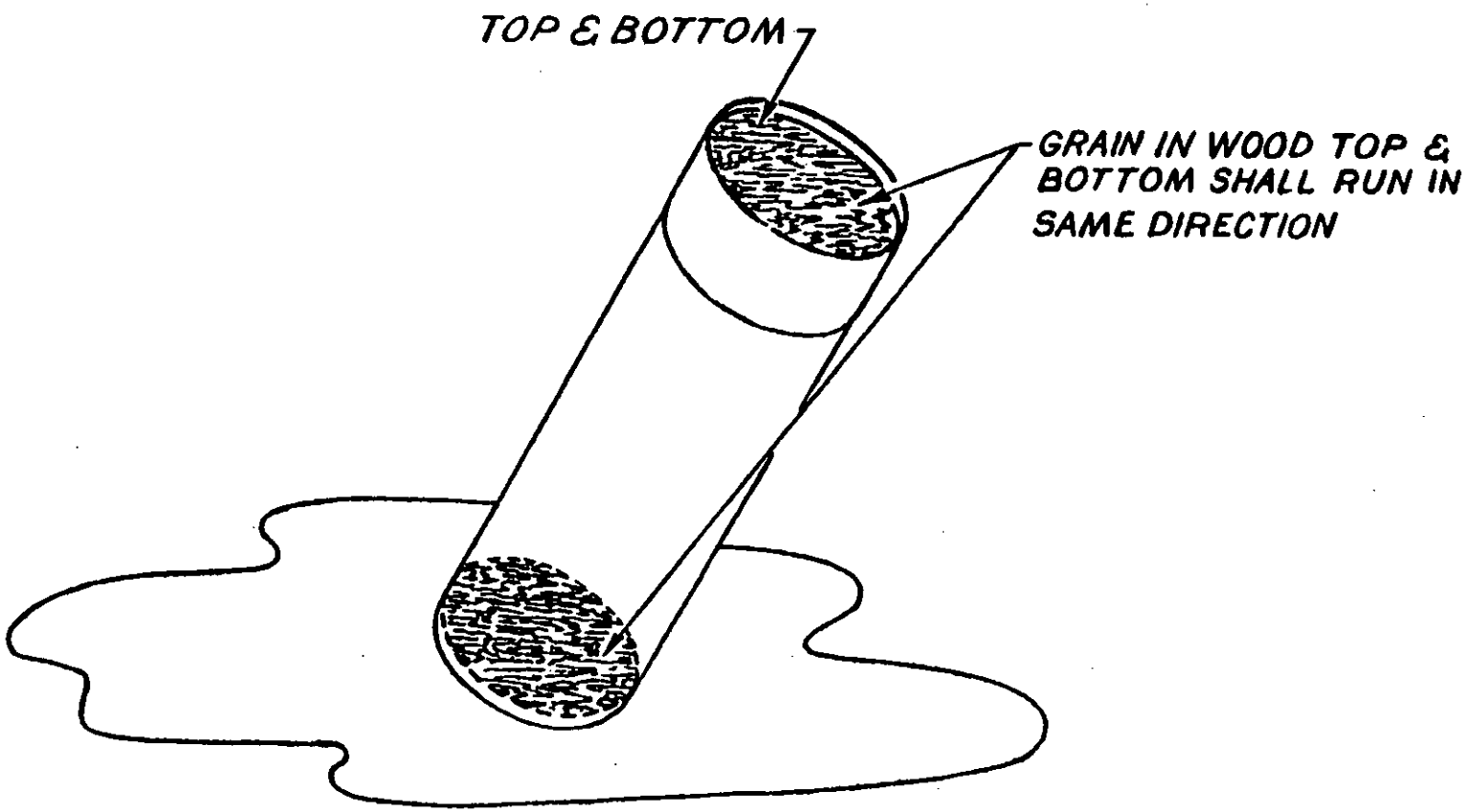
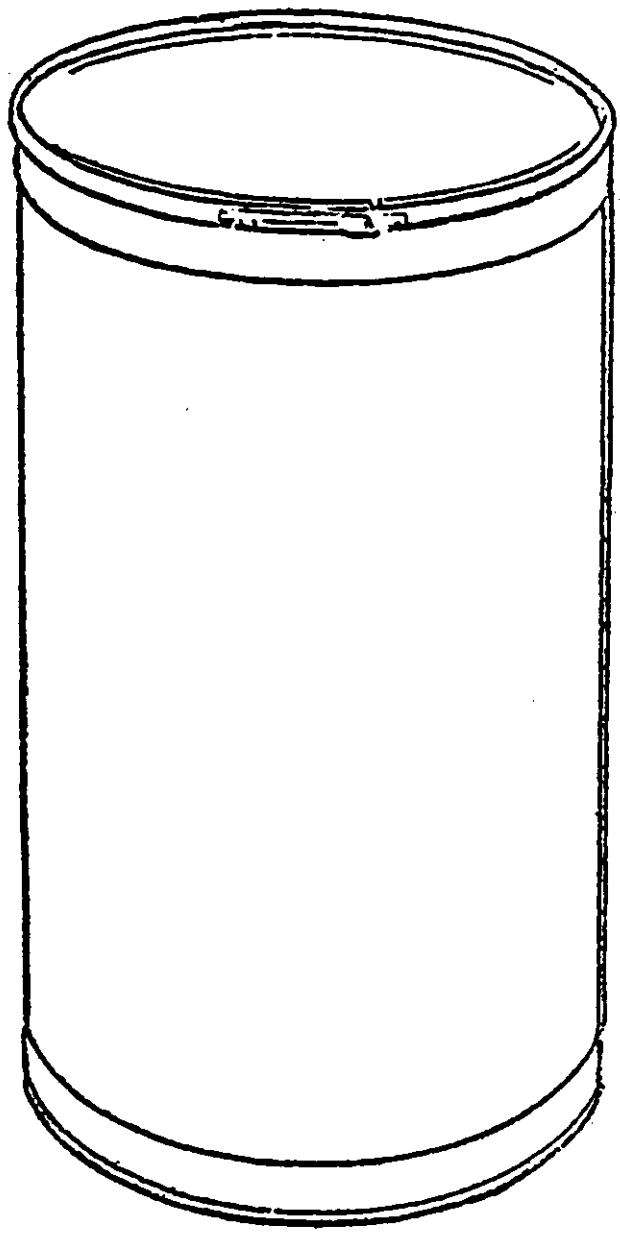


FIGURE 2. Alignment of top and bottom of drums with wood headings.

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METAL LOCKING BAND HOLDS REMOVABLE COVER IN PLACE.

CONVOLUTELY WOUND FIBRE DRUM, HAVING COVER WITH TOP AND BOTTOM CHIME CONSTRUCTION.

FIGURE 3. Fiber drum, chime construction.

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TYPE I

(DOMESTIC TYPE)

(NON-WEATHER RESISTANT)

COMPLIES WITH FED. SPEC. PPP-D-723J

FOR DOMESTIC SHIPMENT

GRADE _____ CLASS _____

MAX. WT. OF CONTENTS _____ LBS.

MAX. CAPACITY CONTENTS _____ GAL.

TYPE II

(OVERSEAS TYPE)

(NON-WEATHER RESISTANT)

COMPLIES WITH FED. SPEC. PPP-D-723J

FOR NORMAL OVERSEAS SHIPMENT

GRADE _____ CLASS _____

MAX. WT. OF CONTENTS _____ LBS.

MAX. CAPACITY CONTENTS _____ GAL.

TYPE III

(WEATHER RESISTANT OVERSEAS TYPE)

COMPLIES WITH FED. SPEC. PPP-D-723J

FOR MILITARY OVERSEAS SHIPMENT

GRADE _____ CLASS _____

MAX. WT. OF CONTENTS _____ LBS.

MAX. CAPACITY CONTENTS ____ GAL.

FIGURE 4. Marking requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
PPP-D-723J	Drums, Fiber

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

NATICK OP, 1 Apr 84

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